THE J-0 PROPERTY DESCENDS ALONG AN INJECTIVE FINITE TYPE MAP OF DOMAINS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- J-0 property of a ring, meaning that the regular locus is open. defined in the same section, source lines 11480-11645, before the counted statement. Supplies the property that is descended.
- Generic flatness over a Noetherian domain for a finite type extension. the author cites Algebra, Lemma algebra-lemma-generic-flatness-Noetherian at proof lines 11660-11662. Makes the extension faithfully flat after localising the target.
- Descent of regularity along a faithfully flat ring map. the author cites Algebra, Lemma algebra-lemma-descent-regular at proof lines 11662-11664. Produces regularity of the base after the second reduction.

Source reference. Original lemma 07PA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9790–9790. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 051R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6544–6544. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9761–9761. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $R \to S$ be a ring map. Assume that*

(1) *R is a Noetherian domain,*
(2) *$R \to S$ is injective and of finite type, and*
(3) *S is a domain and J-0.*

Then R is J-0.

Proof. After replacing S by S_g for some nonzero $g \in S$ we may assume that S is a regular ring. By generic flatness we may assume that also $R \to S$ is faithfully flat, see Algebra, Lemma 051R. Then R is regular by Algebra, Lemma 07NG. □